

Project Report: Metro Ticket Booking System in ServiceNow

Project Title:

Metro Ticket Booking System in ServiceNow

Submitted by:

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GitHub Repository:

<https://github.com/nandanrathodnr/ServiceNow-Project>

Abstract:

This project demonstrates the development of a Metro Ticket Booking System using ServiceNow. It allows users to book metro tickets or recharge metro cards through a Service Catalog item. The system captures user inputs such as source, destination, number of passengers, and journey type. Using Flow Designer, the system automatically calculates fare, generates tickets, and provides QR codes for payment. This automation reduces manual effort, improves accuracy, and enhances user experience.

Introduction

ServiceNow is a cloud-based platform used for workflow automation. This project focuses on automating metro ticket booking.

Manual ticket booking causes:

- Delays
- Errors in fare calculation
- No proper tracking

This system solves the problem by:

- Providing an online booking form
- Automatically calculating fare
- Generating QR-based tickets
- Automating workflow using Flow Designer

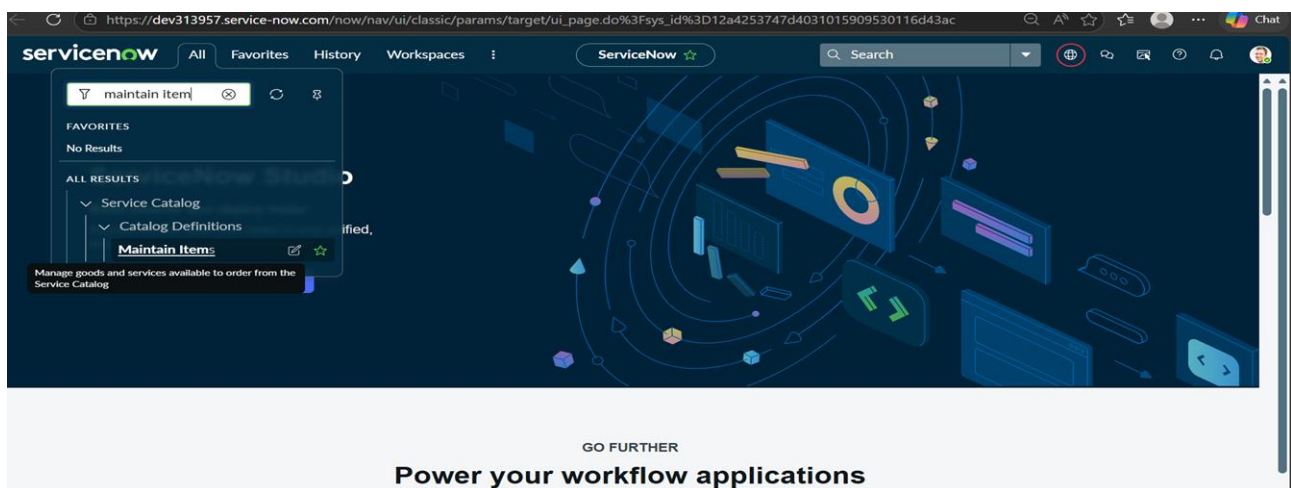
Objectives

1. **Automation** – Eliminate manual ticket booking
2. **Accuracy** – Ensure correct fare calculation
3. **User Experience** – Provide simple booking interface
4. **Efficiency** – Faster booking and processing
5. **Tracking** – Maintain records of all tickets

Methodology

Step 1: Create Catalog Item

- Created "Metro Ticket Booking" in Service Catalog
- Users can select:
 - Book Ticket
 - Recharge Card



Catalog Item
Book A Metro Ticket Item

Manage Attachments (3): metro_logo.png [rename][download] Pasted image.png [rename][download] Pasted image.png [rename][download]

Build and modify items faster with the improved Catalog Builder.

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Book A Metro Ticket Item

Application: Metro Ticket System

Catalogs: Service Catalog

Category: Services

State: -- None --

Checked out: -- None --

Owner: System Administrator

Active: ☒

Fulfillment automation level: Unspecified

Item Details | Process Engine | Picture | Pricing | Portal Settings

Short description: a metro e-ticketing system allows passengers to purchase and use tickets digitally, typically via mobile app or website, eliminating the need.

Description:

Step 2: Create Variables

Variables used:

- Starting Station
- Destination Station
- Number of Passengers
- Journey Type (Single / Return)
- Recharge Amount

Catalog Item
Book A Metro Ticket Item

Catalog item = Book A Metro Ticket Item

Type	Question	Order
Multiple Choice	What do you want to do today?	100
Single Line Text	Enter Smart Card Number	110
Single Line Text	Enter Smart Card Name	120
Single Line Text	Recharge Amount	130
Reference	Starting From	140
Reference	Going To	150
Single Line Text	Number Of Passangers	160
Multiple Choice	Type Of Journey	170
Single Line Text	Amount Of Single Journey	180
Single Line Text	Amount Including Return	190
Multiple Choice	Mode Of Payment	200
Single Line Text	Enter Payment Mode	210
Single Line Text	Enter Card Number	230
HTML	Scan and pay	240

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Step 3: UI Policy

- If "None" → No fields visible
- If "Recharge" → Only recharge field visible

- If "Book Ticket" → Show booking fields

Related Links

Item Diagnostic
Show VA render type
Run Point Scan

Variables (14) Variable Sets **Catalog UI Policies (5)** Catalog Client Scripts (7) Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles

Related Catalog Items Assigned Topics

Order Search Actions on selected rows... New

Catalog item = Book A Metro Ticket Item

☐	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	upi visibility	(empty)		false	true	false	2026-04-27 03:06:10	100
☐	Book QR Ticket	(empty)		false	true	false	2026-04-26 23:47:50	100
☐	Recharge metro card	(empty)		false	true	false	2026-04-27 00:58:15	100
☐	Mode of Payment	(empty)		false	true	false	2026-04-27 00:09:13	100
☐	Field Visibility	(empty)		false	true	false	2026-04-27 00:12:02	100

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Step 4: Client Script (Fare Calculation)

- Script calculates fare dynamically
- Based on:
 - Distance (stations)
 - Number of passengers
 - Journey type

Related Links

Item Diagnostic
Show VA render type
Run Point Scan

Variables (14) Variable Sets Catalog UI Policies (5) **Catalog Client Scripts (7)** Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles

Related Catalog Items Assigned Topics

for text Search Actions on selected rows... New

Catalog item = Book A Metro Ticket Item

☐	Name	Active	Table	View	Type	Updated
☐	recharge	true			onChange	2026-04-27 00:47:31
☐	going	true			onChange	2026-04-27 01:47:49
☐	type	true			onChange	2026-04-27 02:24:11
☐	cardnumber	true			onChange	2026-04-27 00:46:18
☐	upi qr generation	true			onSubmit	2026-04-27 01:50:22
☐	start	true			onChange	2026-04-27 01:51:13
☐	noof	true			onChange	2026-04-27 01:52:10

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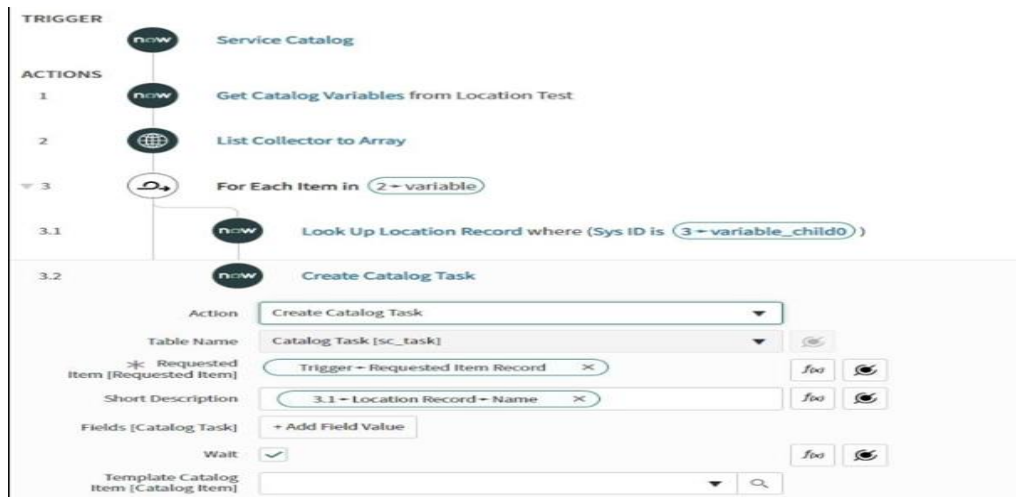
Step 5: Flow Designer

Flow starts when catalog item is submitted.

Actions:

1. Get Catalog Variables
2. Create Ticket Record
3. Calculate Fare
4. Generate QR Code
5. Send Confirmation

Step 6: Code



QR

Generation

- QR code created for payment
- User scans and pays

The screenshot shows the 'Variable Scan and pay' configuration interface. The top section includes fields for 'Application' (Metro Ticket System), 'Type' (HTML), 'Catalog Item' (Book A Metro Ticket Item), and 'Order' (240). There are also checkboxes for 'Active', 'Mandatory', 'Read only', 'Hidden', and 'Disable automatic slot fill based on user context'. The bottom section, titled 'Default Value', shows a 'Default HTML' editor with a QR code and the text 'Scan & Pay' and 'Scan this QR and complete payment'.

Step 7: Notification

- Email sent to user with:
 - Ticket details
 - QR Code

- Fare amount

Item		Requested for	Quantity
Book A Metro Ticket Item		System Administrator	1
Stage			
▶ Assess or Scope Task			

Activity

Attachments

Additional Details

What do you want to do today?
Book QR Ticket

Starting From
Ballari

Going To
Raychur

Number Of Passangers
3

Type Of Journey
Single

Amount Of Single Journey
180

Mode Of Payment
Other

Enter Payment Mode
Cash

Results

1. Users can book metro tickets easily
2. Fare is calculated automatically
3. QR-based payment system implemented
4. Manual errors reduced
5. Faster ticket processing

Conclusion

The Metro Ticket Booking System successfully automates ticket booking using ServiceNow. It improves efficiency, accuracy, and user experience. The integration of QR code payment and automated workflows ensures a smooth and modern booking process.

Future Enhancements

1. Live Metro Tracking
2. Mobile App Integration
3. Dashboard for Admin
4. AI-based Travel Suggestions

